

Analyzing the Co-Movement of U.S. Electricity Price Shocks and Global Macroeconomic Indicators During the **2022-2024 Volatility Period**

Implementation of a Data Warehouse and Business Intelligence System

Group Members:

- Abdullah Al-Firdaus Nuzula (008)
- Fio Ulaa' Octriyanti (030)
- Mylovia Mahesa Ayu (001)
- Muhammad Raffi Fahrezi (100)

Background

Introduction



This project analyzes the relationship between U.S. energy consumption and global macroeconomic factors using a data warehouse approach.

1

The Russia-Ukraine and Middle East conflicts disrupted global energy and economic stability.

These geopolitical events increased uncertainty in energy supply chains and global market conditions.

2

Electricity consumption is closely related to industrial activity, inflation, and economic growth.

Changes in macroeconomic conditions directly influence electricity demand and pricing behavior.

3

Global economic indicators help measure the impact of economic conditions on energy demand.

Indicators such as GDP and CPI provide insights into worldwide economic and energy trends.

4

The 2022-2024 period was highly volatile due to post-pandemic recovery, inflation, and geopolitical crises.

As a result, both electricity markets and global economic performance experienced significant fluctuations.

Research Objective

Volatility Periods



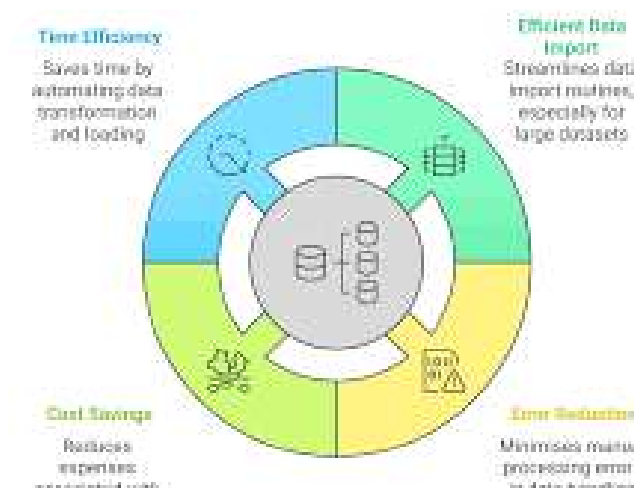
Analyze U.S. electricity price and consumption trends during the 2022-2024 volatility period. This analysis aims to identify fluctuations in the energy sector during global economic instability.

Economic Correlation



Examine the relationship between electricity prices and global macroeconomic indicators. The study investigates how GDP, CPI, and industrial production interact with energy market dynamics.

ETL Automation



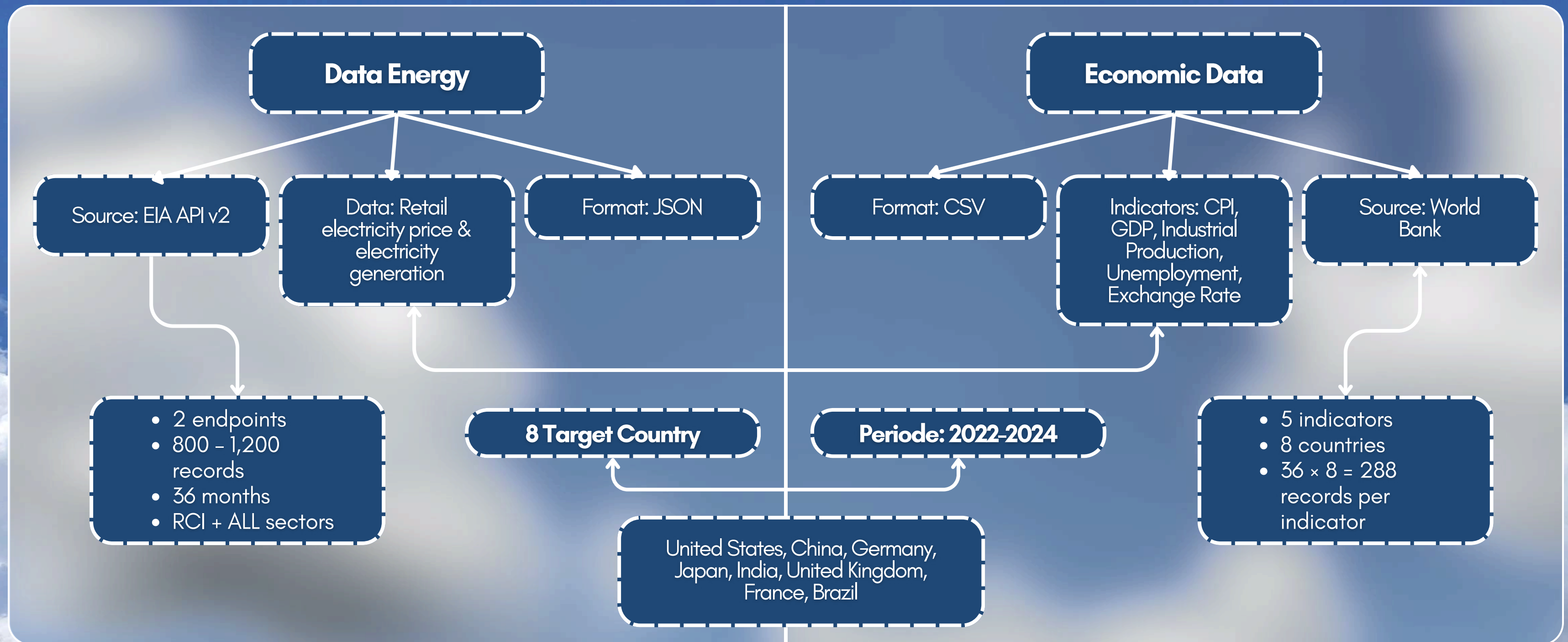
Develop an automated ETL pipeline using Apache Airflow and PostgreSQL. The pipeline enables continuous extraction, transformation, and loading of heterogeneous datasets.

Star Schema



Implement an OLAP-ready star schema for multidimensional analysis using Atoti. This structure supports efficient business intelligence and analytical querying processes.

Dataset Source Identification



Data Characteristics & Anomalies

Sparsity

Unemployment data has a **25%** sparse rate, especially absent in India & China.

Imbalance

The volume of **natural gas supply** dominates significantly, up to **3–4 times higher** than other types of fuel.

Outliers

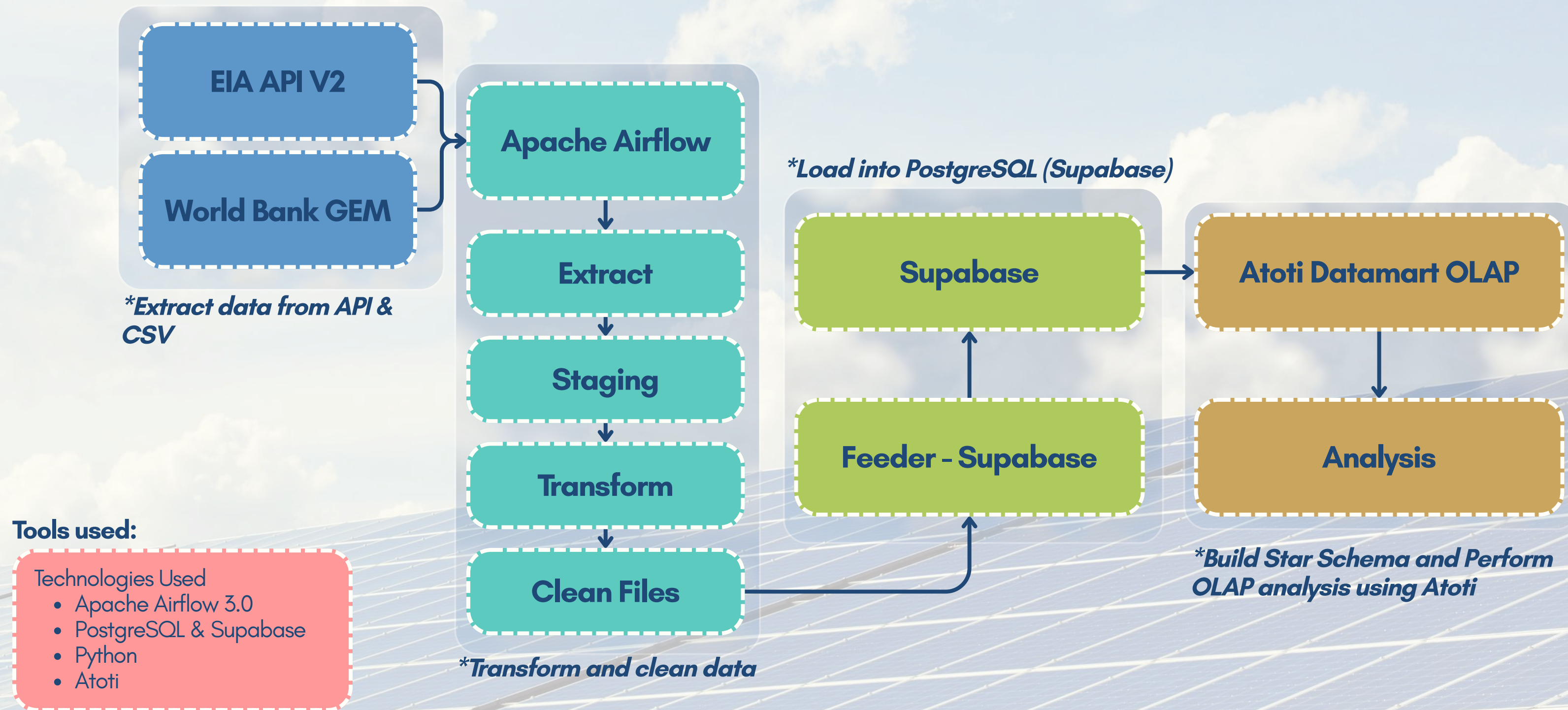
The United States CPI jumped to **9.1% in June 2022**. This was identified as a genuine data point, not an error.



GDP Forward-Fill Transformation

Transforms rigid annual GDP visual data into consistently distributed 12-month granular data.

End to End Data Warehouse System



4 Research Hypotheses

Hypothesis	Theoretical Basis	Initial Expectation
H1: U.S. industrial electricity price surge co-moves with G7 industrial production	electricity price surge co-moves with G7 industrial production Energy price transmission theory: rising energy input costs in a major economy propagate to industrial output of trade-linked economies through global supply chains	Significant negative contemporaneous correlation between U.S. industrial electricity prices and G7 industrial production indices, with delayed effects detectable at lag 1-3 months
H2: U.S. retail electricity prices correlate with global CPI	Cost-push inflation theory: energy as a universal production input – when electricity prices rise, production costs increase across sectors, pushing consumer prices upward	Significant positive correlation between U.S. retail electricity prices and CPI across eight target economies (United States, Germany, Japan, United Kingdom, France, Brazil, China, India)

[Ganepola et al. \(2023\)](#) [Garzon & Hierro \(2024\)](#) [Alvarez et al. IMF \(2024\)](#) [Khalfaoui et al. \(2023\)](#)

4 Research Hypotheses

Hypothesis	Theoretical Basis	Initial Expectation
H3: Higher U.S. renewable energy share reduces CPI volatility	Renewable insulation hypothesis: renewables have near-zero marginal fuel costs, making economies with higher renewable penetration more resilient to fossil fuel price shocks that drive inflation volatility	Significant negative correlation between U.S. renewable generation share and CPI volatility (measured as 6-month rolling standard deviation)
H4: June–August 2022 constitutes a simultaneous multi-indicator anomaly period	Systemic shock theory: converging crises (post-pandemic recovery, Russia-Ukraine conflict, energy market volatility) are expected to produce synchronized extreme deviations across all macroeconomic and energy indicators	All key indicators breach the $ Z > 2.0$ threshold simultaneously during June–August 2022, confirming a "perfect storm" anomaly window

[Ganepola et al. \(2023\)](#) [Garzon & Hierro \(2024\)](#) [Alvarez et al. IMF \(2024\)](#) [Khalfaoui et al. \(2023\)](#)

ETL Detail: Extract

Extract phase • Apache Airflow 3.0 PythonOperator

EIA API V2

Mechanism

(a) REST API; (b) requests.get(); (c) format: JSON

Endpoints

/electricity/retail-sales/data
/electricity/electric-power-operational-data

Filter Params

location=US • frequency=monthly • 2022-2024

Output

800 – 1,200 records; 36 months; /staging/

World Bank GEM

Mechanism

(a) REST API; (b) requests.get(); (c) format: JSON

Indicators

CPI • GDP • Industrial Production
Unemployment Rate • Exchange Rate

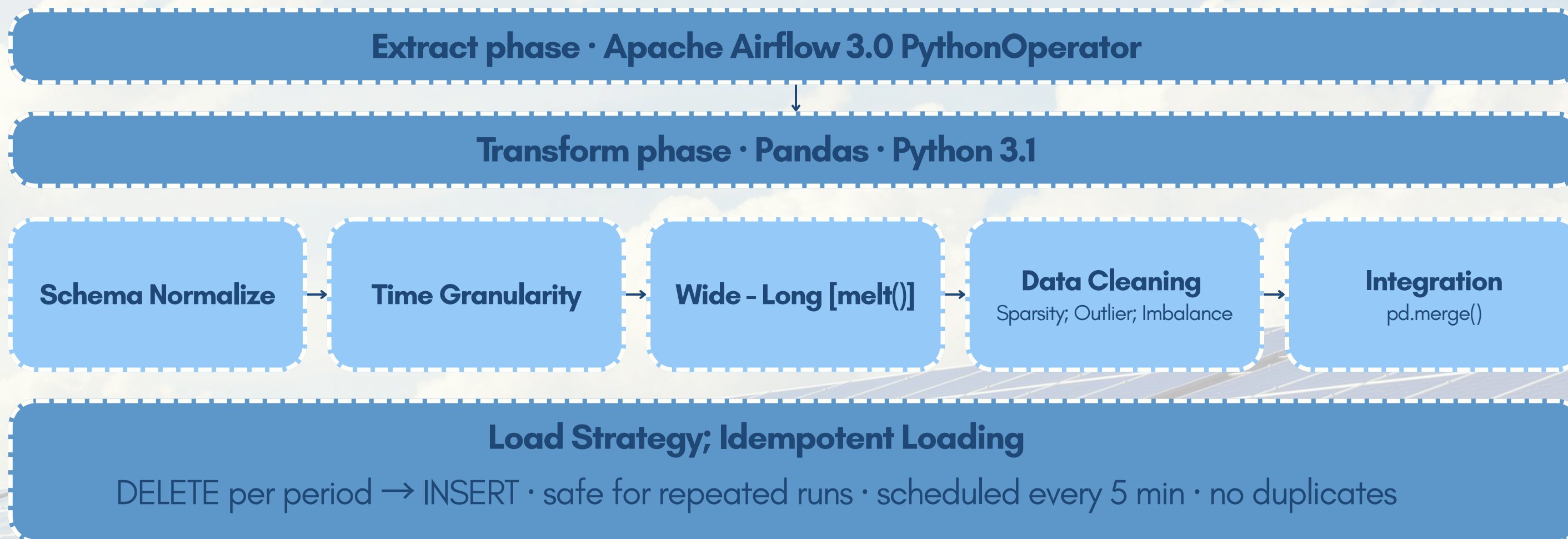
Coverage

8 countries • 36 periods × 8 = 288 rows/indicator

Output

5 CSV files; /staging/; no API dependency

ETL Detail: Transform

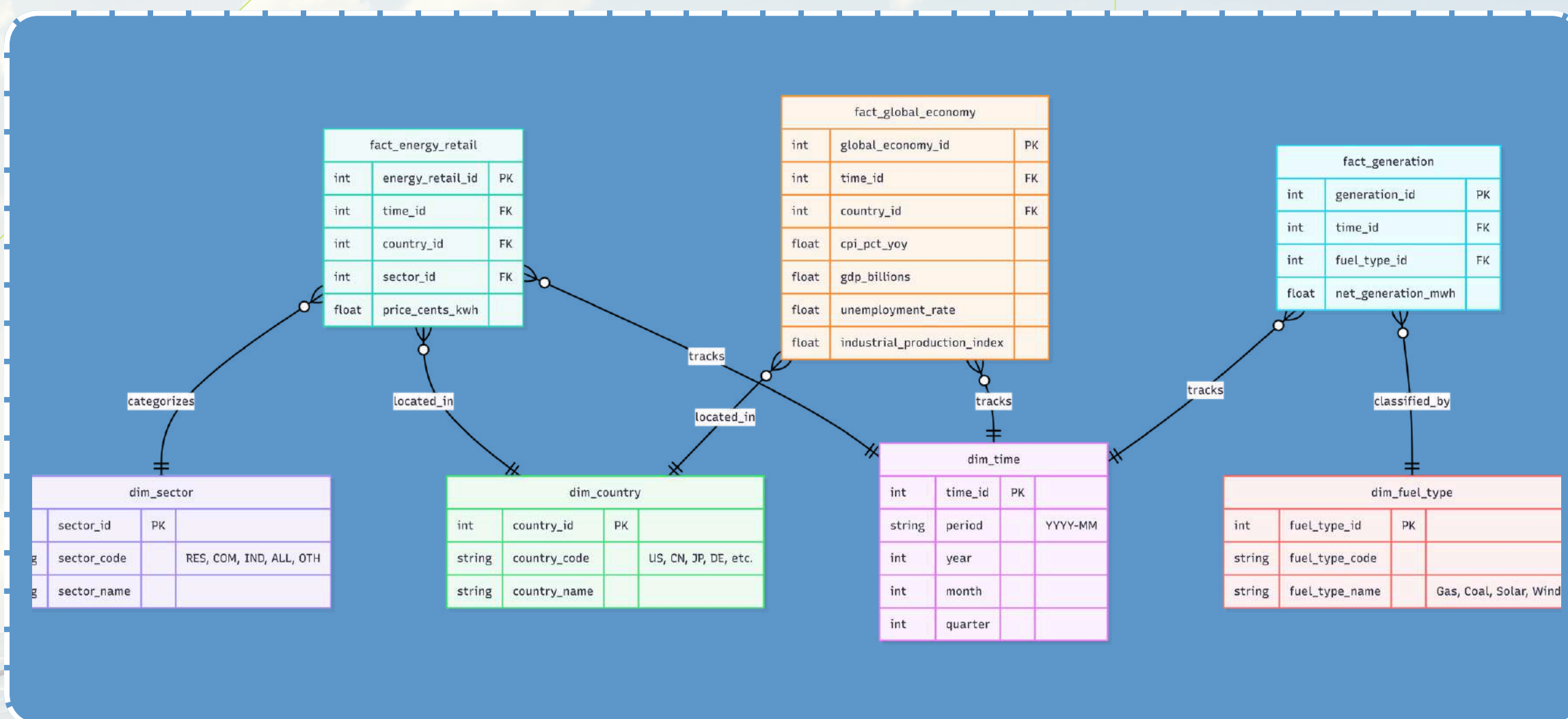


Statistical Method

Method	Purpose	Library	Hypothesis
Pearson correlation	Measures linear relationship between two continuous variable	scipy.stats	H1, H2, H3
Spearman correlation	Measures monotonic (non-linear) relationship, rank-based	scipy.stats	H1, H2
Cross-correlation lag 0-3	Detects delayed transmission of price shocks over 1-3 months	scipy.stanumpy · scipy.correlatets	H1
Rolling std 6-month	Measures CPI volatility as 6-month rolling standard deviation	pandas	H3
Z-score Standarization	Detects simultaneous anomalies across indicators · threshold $ Z > 2.0$	scipy.stats	H4

Data Modeling

Star Schema Design



- This model optimizes queries by placing **numerical metrics at the center** (Facts) and **descriptive attributes around them** (Dimensions).
- **3 Fact Table:** fact_energy_economy, fact_worldbank & fact_generation
- **4 Dimension Tables:** Time, Fuel Type, Sector, Country.
- Data Grain Level:
 - **fact_energy_economy** (1 row = 1 Sector × 1 Month)
 - **fact_generation** (1 row = 1 Fuel Type × 1 Month)

Dimension Tables Details

Dim_Time

Year, Quarter, Month.

Dim_Fuel_Type

Coal, Gas, Nuclear, Renewable

Dim_Sector

Residential, Industrial, Commercial,
All Sectors, Others

Dim_Country

Country Code, Country Name,
World Region, Income



Fact Table

fact_energy_economy

Metric Type	Column Name	Description
Key	fact_id	Surrogate primary key
Key (FK)	time_id	→ dim_time
Key (FK)	sector_id	→ dim_sector
Market	price_cents_kwh	Retail electricity price per kWh
Energy	retail_sales_mwh	Total electricity sold (MWh)



Fact Table

fact_energy_economy

Metric Type	Column Name	Description
Market	revenue_million_usd	Revenue from electricity sales
Inflation	cpi_pct_yoy	Consumer Price Index (YoY %)
Economy	gdp_current_usd_mn	GDP in million USD
Economy	industrial_prod_usd	Industrial Production index
Economy	unemployment_rate	Unemployment rate (%)
Economy	exchange_rate_lcu_usd	Exchange rate LCU per USD

Fact Table

fact_generation

Metric Type	Column Name	Type	Description
Key (PK)	gen_id	int	Surrogate primary key
Key (FK)	time_id	int	→ dim_time
Key (FK)	fuel_type_id	int	→ dim_fuel_type
Attribute	period	str	Format YYYY-MM
Attribute	fuel_name	str	Coal, Wind, Solar, dst
Attribute	fuel_category	str	Fossil / Renewable / Nuclear / Other
Attribute	is_renewable	bool	True / False
Measure	net_generation_mwh	flat	Satu-satunya measure — fully additive

Grain: 1 row = 1 Fuel Type × 1 Month → 288 rows (8 fuel types × 36 bulan)



Fact Table

fact_worldbank_indicators

Metric Type	Column Name	Description
Key (PK)	period	Month identifier: format YYYY-MM (e.g. 2022-06)
Key (FK)	country_code	ISO country code (US, CN, DE, JP, IN, GB, FR, BR)
Time	country_name	Full country name (e.g. United States, Germany)
Time	year	Year extracted from period (2022, 2023, 2024)
Time	quarter	Quarter number (1-4) for ROLLUP aggregation
Inflation	month	Month number (1-12)



Fact Table

fact_worldbank_indicators

Metric Type	Column Name	Description
Economy	cpi_pct_yoy	Consumer Price Index; year-over-year % change
Economy	gdp_current_usd_mn	GDP at market prices in millions USD
Economy	industrial_prod_usd	Industrial Production index (constant 2010 USD)
Economy	unemployment_rate	Unemployment rate; seasonally adjusted (%)
Economy	exchange_rate_lcu_usd	Exchange rate; local currency units per 1 USD

PostgreSQL Optimizations

4 Features

Partitioning

By Year: Segmentation of physical data storage based on the period year.

B-Tree Index

Target: period & fuel_id.
Speeds up log search & dimension relations.

M-Views

mv_generation_monthly.
Complex aggregations are stored physically on disk.

Monitoring

pg_stat_statements.
Tracks query performance for bottleneck detection.

432

Fact table rows

55

Dimension rows

Note: Performance benchmark testing is planned as a follow-up agenda.

OLAP Insight



This project analyzes the relationship between U.S. energy consumption and global macroeconomic factors using a data warehouse approach.

1

Time Series Decomposition

showing the monthly and quarterly trajectory of U.S. net electricity generation by fuel type across 2022-2024. Based on general energy transition knowledge, the expected pattern is a gradual decline in coal generation, stable nuclear generation, and increasing wind and solar generation

2

Price Sensitivity Matrix

cross-tabulating average electricity price changes by sector against CPI movements by country, with the hypothesis that U.S. residential price increases in 2022 correlate positively with CPI spikes in other major economies

3

Generation-Demand Alignment Analysis

Examining whether quarterly industrial electricity consumption in the U.S. leads, lags, or coincides with industrial production index movements in target economies, providing evidence for electricity consumption as a real-time economic activity indicator.

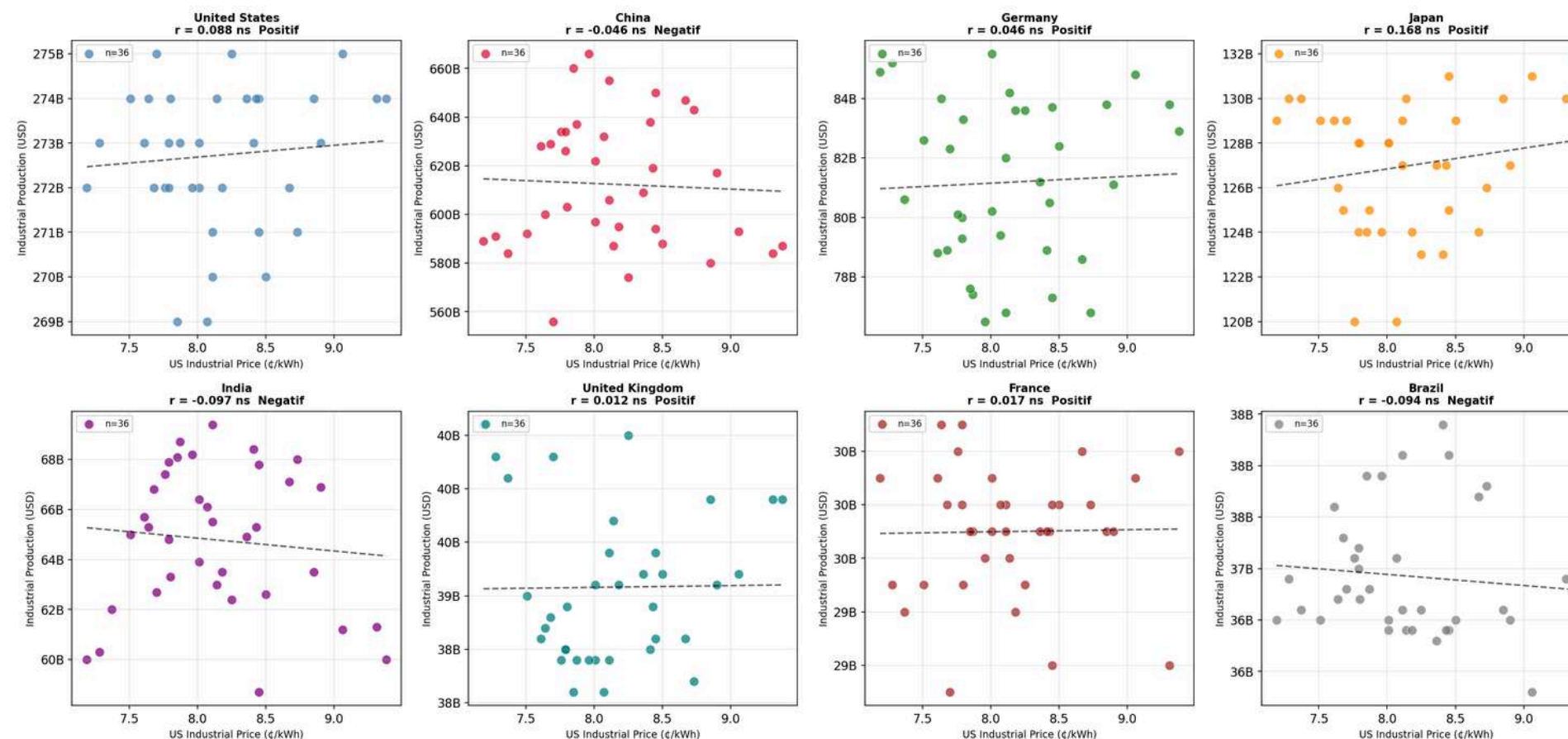
4

Renewable Transition Velocity

CUBE rollup queries computing the renewable energy percentage share (wind + solar + hydroelectric as proportion of total generation) per quarter, quantifying the rate of energy mix transition across the three-year observation window.

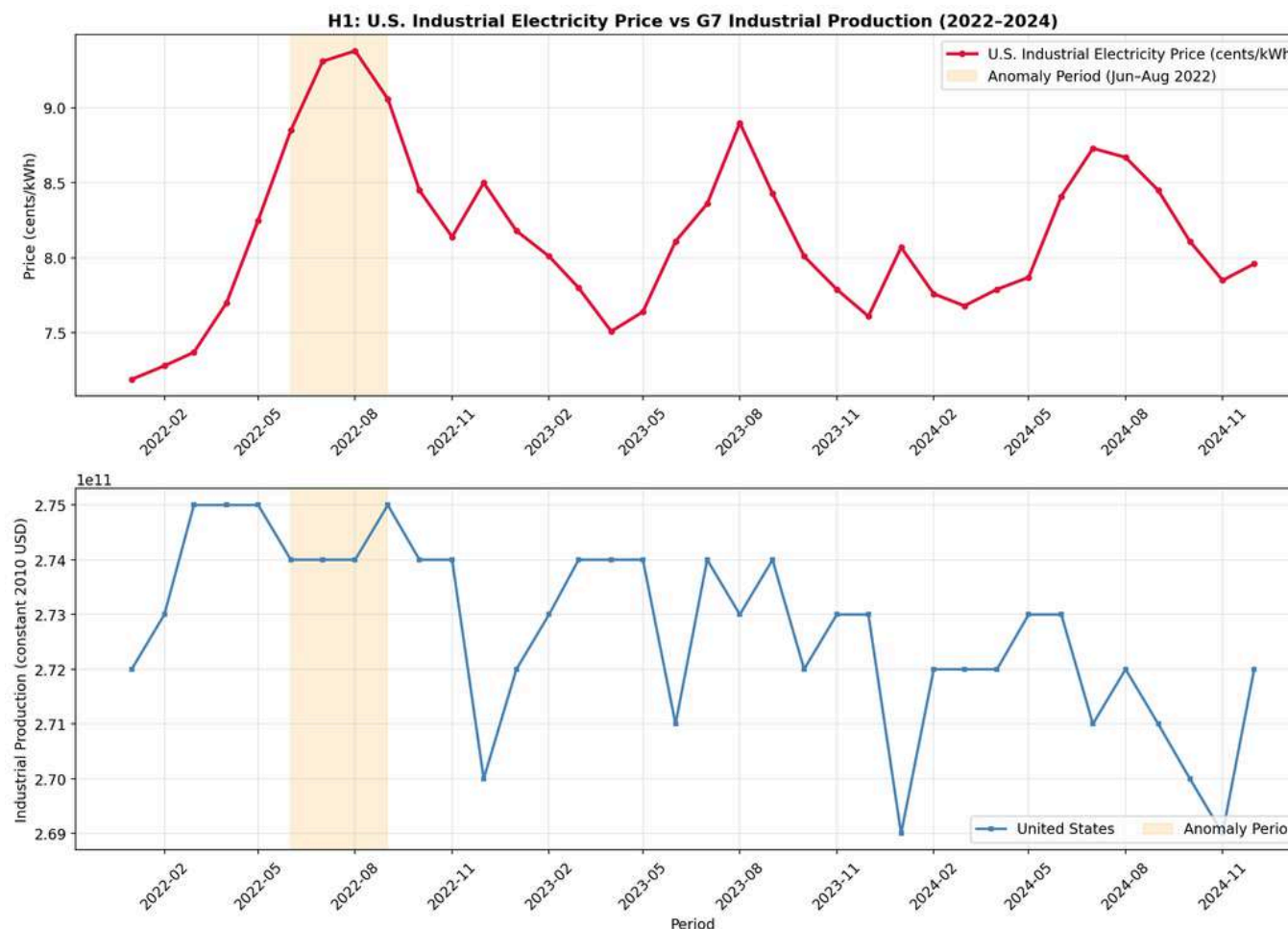
H1: Industrial Output vs. Energy Pricing

H1: Scatter Plot - U.S. Industrial Electricity Price vs Industrial Production per Country (2022-2024)



- **Core Finding:** Industrial production cycles have a statistically **non-significant relationship** with U.S. retail electricity pricing.
- **Weak Linear Correlations (r):**
 - U.S.: +0.088 (weak positive)
 - China: -0.046 (weak negative)
 - India: -0.097 (weak negative)
- Key Takeaway: Fluctuations in national manufacturing output **do not drive U.S. industrial energy costs.**

H1: Mid-2022 Price Surge vs. Production



- The 2022 Anomaly (June–Aug): U.S. Industrial Electricity Prices spiked sharply, **peaking at >9.0 cents/kWh.**
- Industrial Output: U.S. production plateaued during this exact price peak, then dipped later.
- Core Takeaway: **This massive disconnect proves energy volatility** was driven by external global shocks, not local industrial demand.
- Global Drivers: **Post-pandemic** recovery and the **Russia-Ukraine** conflict.

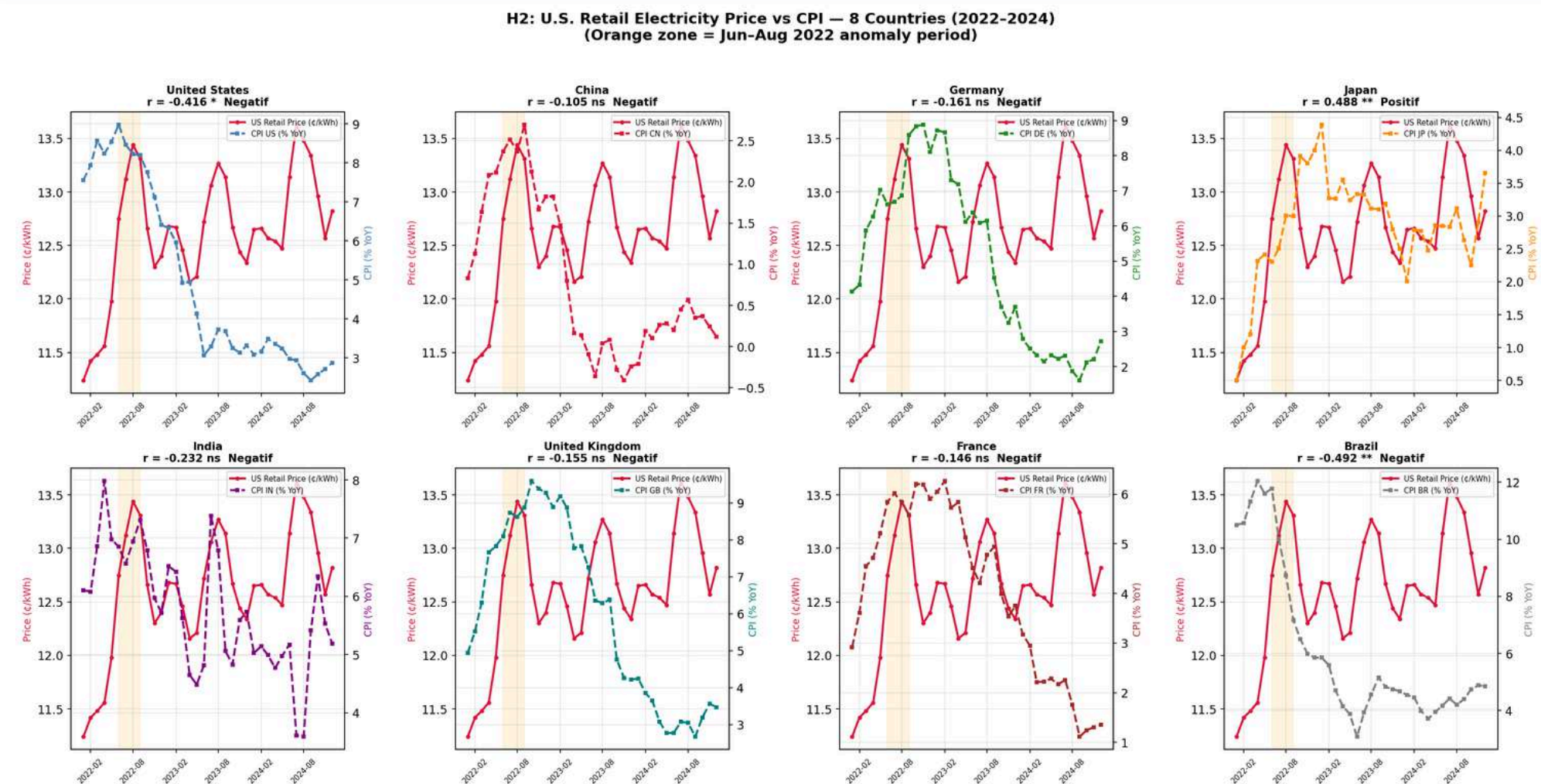
H2: U.S Electricity Pricing vs. CPI Correlation

- The Problem: A static **correlation coefficient** hides the truth.
- The Reality: **The 6-month rolling r reveals** extreme volatility.
- The Rollercoaster: Strong Positive ($r > 0.75$) → Zero → Strong Negative ($r < -0.5$) → Sharp Spike
- Conclusion: The economic relationship between **energy pricing and inflation** is **inherently unstable**.

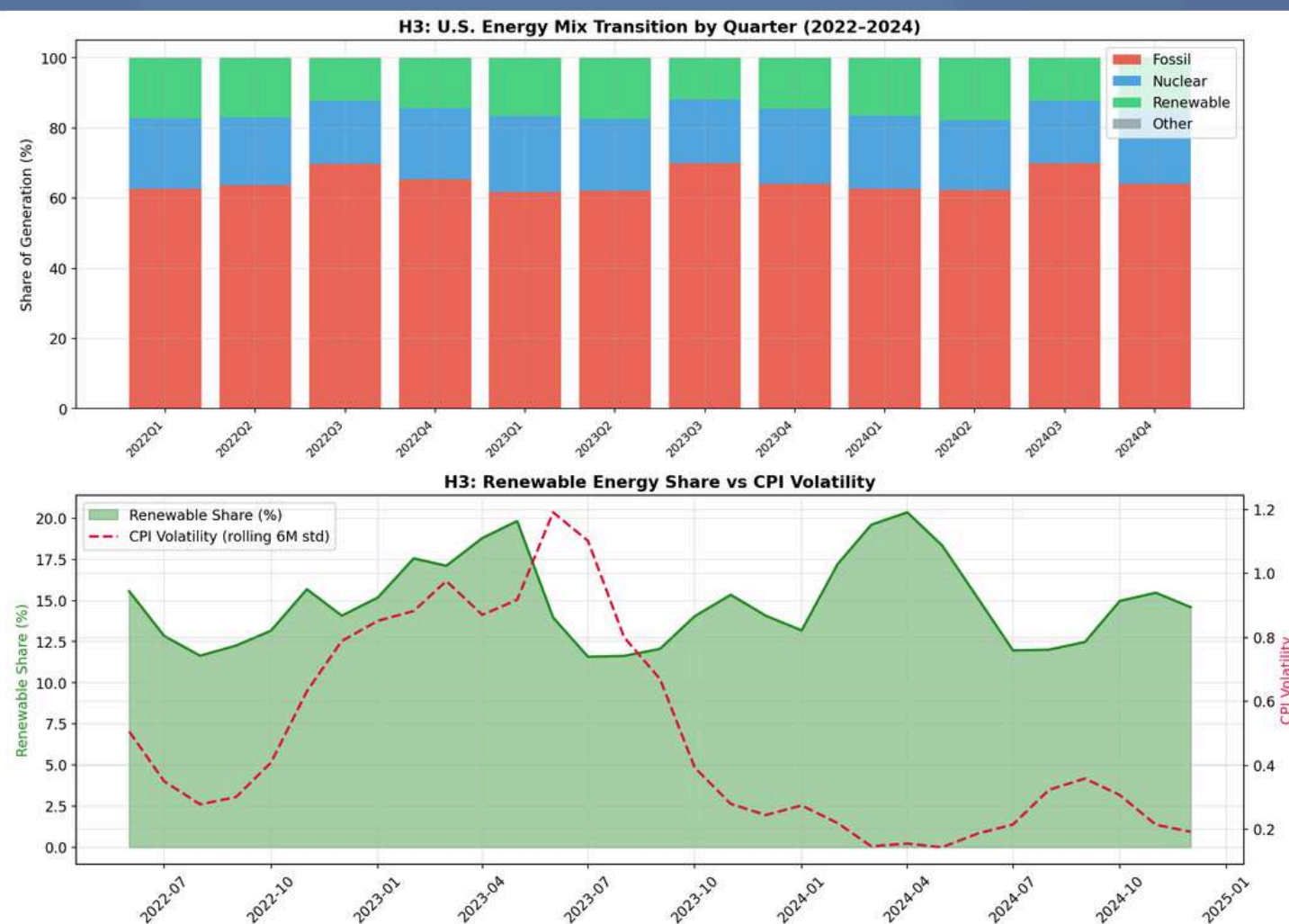


H2: Global Correlation of Energy Prices vs. CPI

- **Core Finding:** The relationship between electricity prices and inflation (CPI) varies drastically by country—there is no uniform global trend.
- **The 2022 U.S. Anomaly (June–Aug):** Peak U.S. inflation (8–9% y-o-y) coincided with a **steep upward trajectory in electricity prices**, resulting in an overall moderate negative correlation ($r = -0.416$).
- **Global Divergence (r):**
 - Brazil: Notable negative correlation (-0.492)
 - Japan: Moderate positive correlation ($+0.488$)



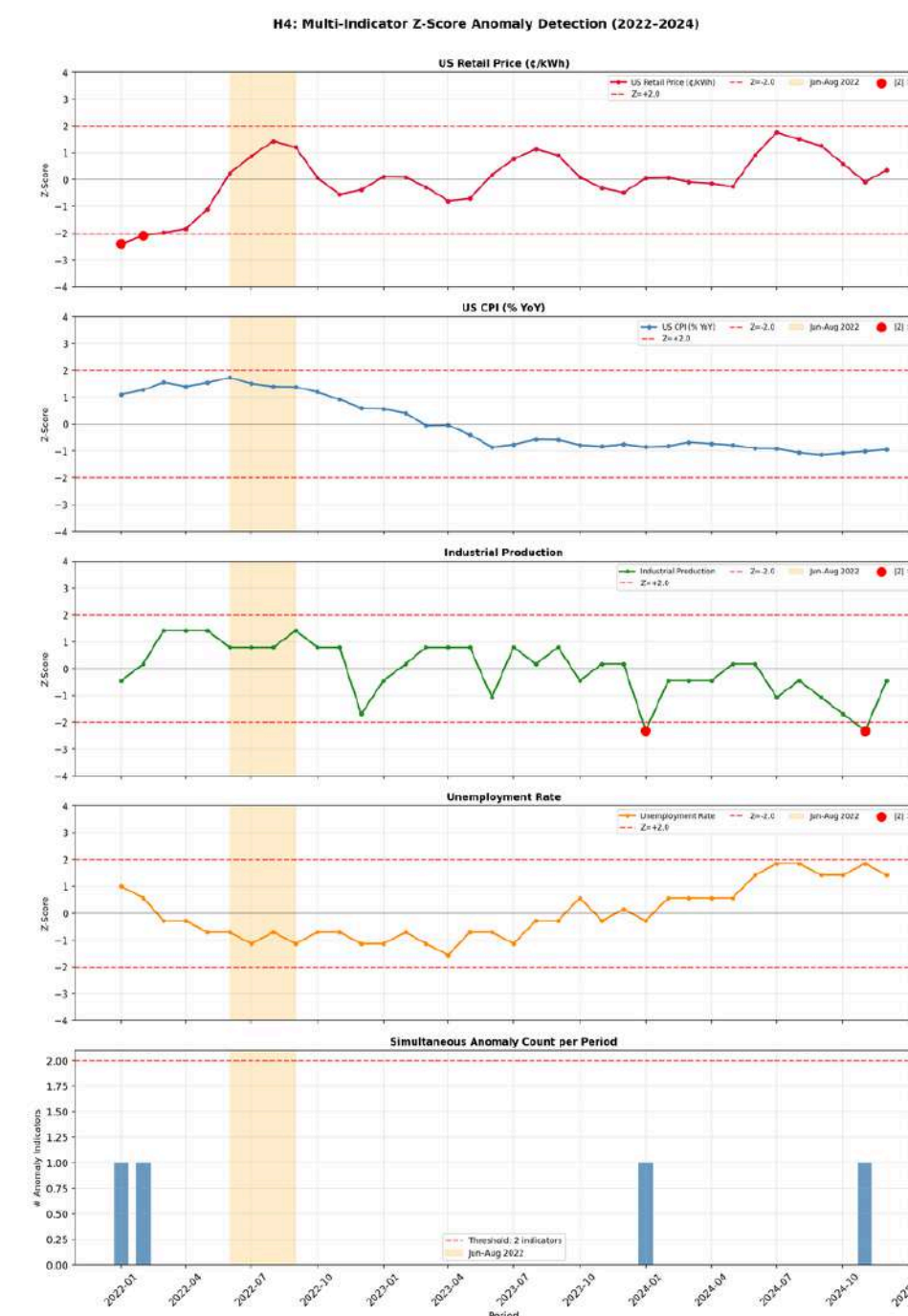
H3: Generation Shifts & CPI Volatility



- **The Seasonal Tug-of-War:** U.S. generation relies on a predictable, cyclical flip-flop:
 - Summer (Q3): Fossil fuels peak (60%–70% share) to meet cooling demand.
 - Spring (Q2): Renewables hit their stride, peaking near 20%.
- **Key Correlation:** Early-to-mid 2023 saw peak inflation volatility happen at the exact same time the renewable transition share was surging toward its local maximum.

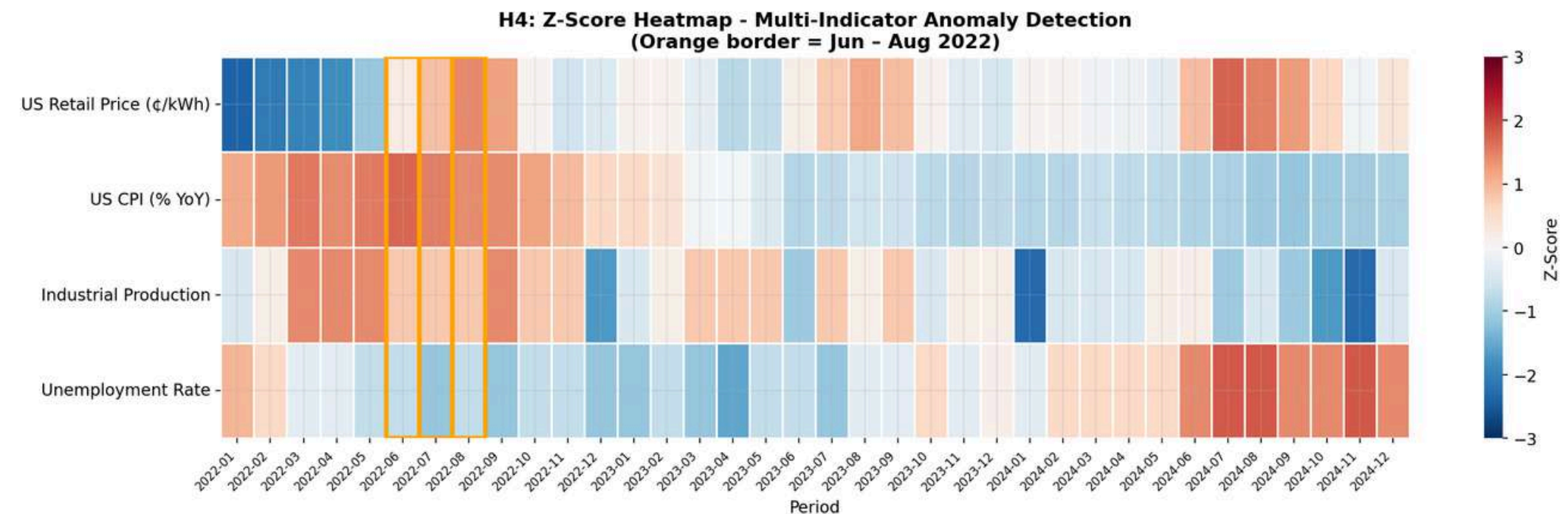
H4: Shock Transmission (Synchronized vs. Domino Effect)

- **The Core Finding:** The transmission of macroeconomic shocks into the energy sector was **a temporally distributed domino effect**, not a single synchronized collapse.
- **The 2022 Window Reality:** While highly volatile, individual indicators (CPI vs. Retail Prices) peaked at lagging intervals.
 - Concurrent monthly breaches **never crossed the threshold of ≥ 2 variables** simultaneously.
- **Isolated Extremes:** True simultaneous breaches happened at independent timestamps:
 - Early 2022: Depressed retail prices.
 - Late 2024: Sharp contraction in the U.S. Industrial Production Index.



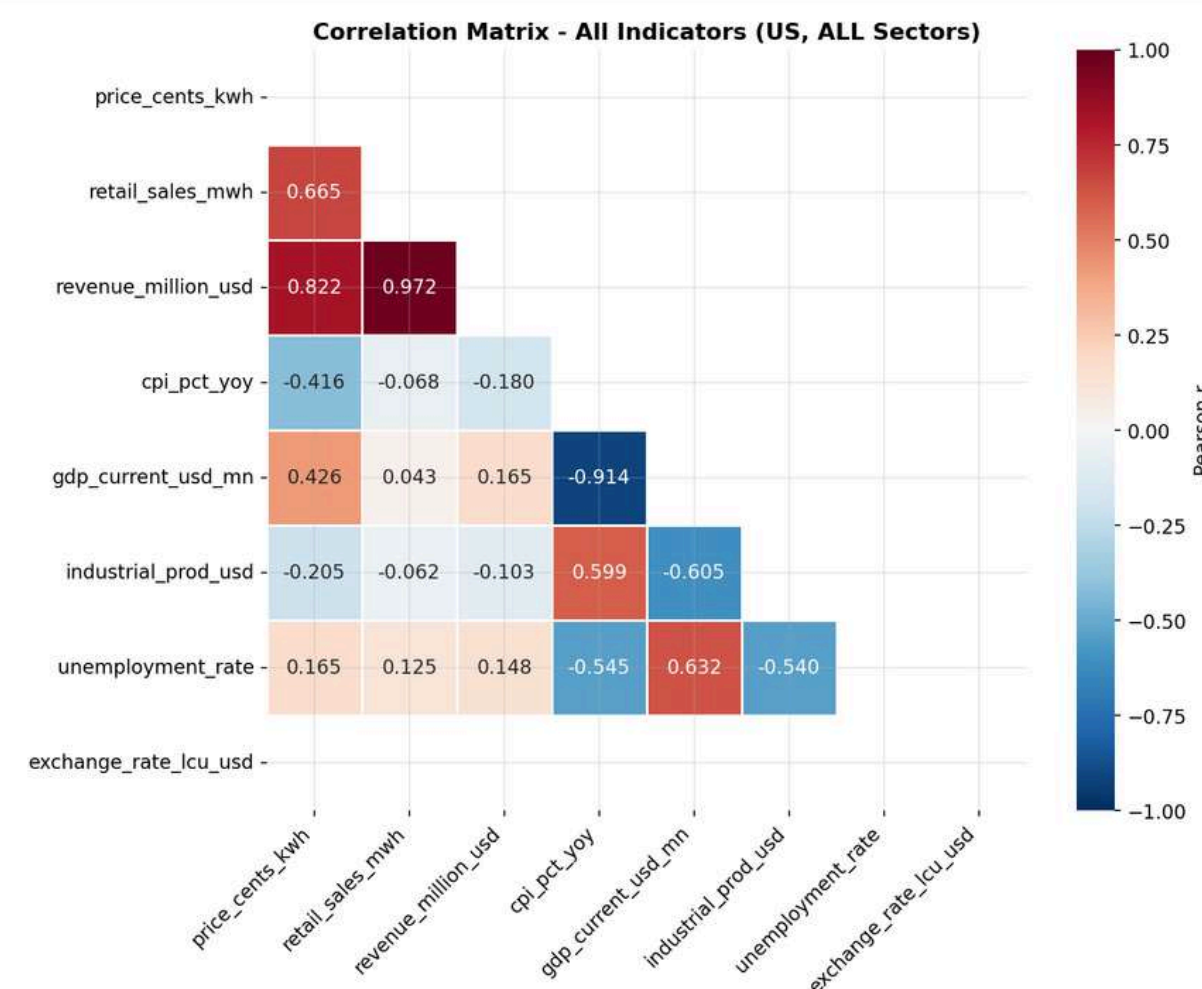
H4: Z-Score Heatmap - Multi-Indicator Anomaly Detection

- **The Asset:** A pairwise Pearson correlation matrix (r) mapping key energy and macroeconomic indicators.
- **Visual Legend:**
 - Red Hues: Positive linear relationships ($r \rightarrow +1.0$).
 - Blue Hues: Negative linear relationships ($r \rightarrow -1.0$).
 - Intensity: Reflects the statistical strength of the correlation.
- **Core Function:** Instantly isolates which economic drivers **directly move alongside energy metrics** and which operate independently ($r \approx 0$).



Correlation Matrix Overview

- **The Target Window:** Multidimensional analysis isolates June–August 2022 as the primary anomaly period (marked in orange).
- **U.S. CPI Peak:** Deep red saturation indicates historic inflation peaking near the upper historical boundary ($Z \approx +2.0$).
- **Utility Pricing Response:** Retail electricity prices **shifted positively ($Z > +1.0$)** during the same quarter.
- **Core Takeaway:** Macroeconomic inflation pressures translated immediately into utility pricing volatility.



Summary of Hypothesis

H	Hypothesis	Expected	Findings	Verdict
H1	Industrial price $\leftrightarrow$ G7 industrial production, negative correlation	Significant negative correlation, lag 1-3mo	Weak, non-significant, price spike driven by external shocks and not production	Rejected
H2	Electricity price correlates with global CPI	Stable positive correlation	Relationship exists but is unstable — 6-month rolling r swings from +0.75 to -0.5; varies by country	Partially supported / nuanced
H3	Higher renewable share reduces CPI volatility	Significant negative correlation	Renewable peaks (Q2 ~20%) coincided with peak inflation volatility, not reduced it	Rejected (or inconclusive)
H4	Jun-Aug 2022 = simultaneous anomaly across all indicators	≥ 2 indicators breach $Z > 2$ at once	Breaches were temporally distributed, never ≥ 2 simultaneously; true extremes at independent timestamps	Rejected

Conclusion

Project Contributions & Research Value

What We Built

- **End-to-End Reproducible Pipeline Automated ETL via Apache Airflow 3.0:** scheduled every 5 minutes, idempotent loading, 7 staging files → 3 fact tables in Supabase PostgreSQL cloud.
- **OLAP-Ready Star Schema 2 fact tables (432 rows) + 4 dimensions (55 rows)** with B-tree indexing, year partitioning, 2 materialized views, and pg_stat_statements monitoring.
- **Integrated Multi-Source Dataset 36 months** of EIA API data (electricity) × World Bank GEM data (5 macroeconomic indicators × 8 countries), heterogeneous sources unified into one analytical system.

What This Contributes:

- **Empirical Evidence Challenges** the mainstream assumption that **electricity prices are positive co-movers with inflation**. The **2022-2024** data shows the relationship is **unstable, country-specific, and direction-reversed**.
- **A Methodological Template** Demonstrates a replicable **DWH + OLAP + statistical hypothesis** testing pipeline for energy-economy research, extendable to other countries, time periods, or energy sources.

Limitations & Future Work

Limitations

- **GDP forward-fill.** Annual monthly forward-fill injects artificial flatness into a key indicator, weakening any monthly correlation involving GDP.
- **Sparsity.** Unemployment missing for India & China (25%); "G7" labeling is inconsistent with your actual 8-country set.
- **No causal inference.** Pearson/Spearman/lag show association only.

Future Work

- **Extend the time span** (pre-2022 baseline) for more statistical power.
- **Add state/regional granularity** for more granularity and analytical power.
- **Interpolate GDP better** (cubic spline) or **source a monthly proxy**.

Any Questions?

Thank You

"Decoding the Pulse of Nations, where energy meets the economy."

[GitHub Repository](#)

